

FEVengers · Doctor Whodunit

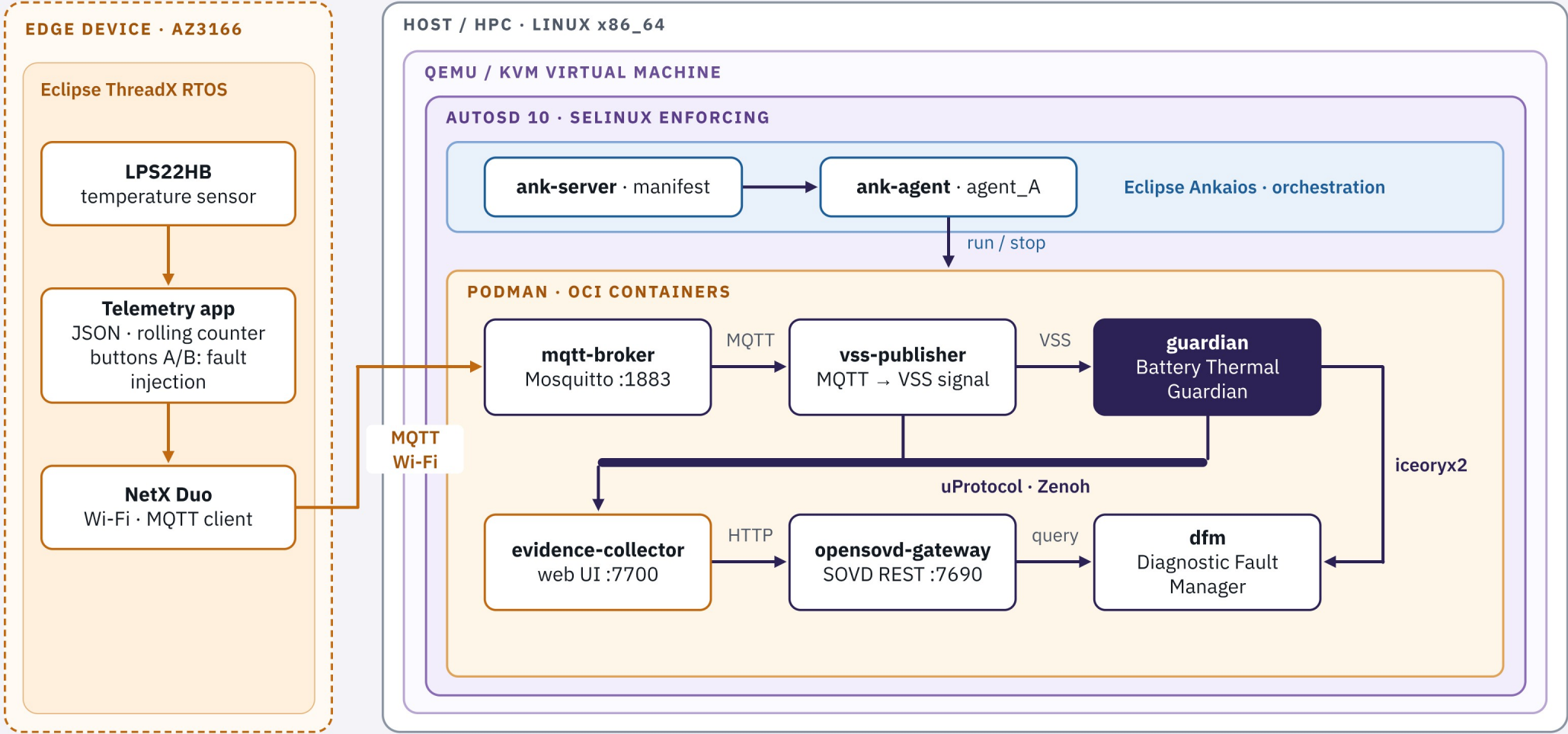
Battery Thermal Guardian – a safety evidence factory on AutoSD. Every fault leaves evidence.

TEAM · FEV.io × FEV TÜRKIYE

Ashwin Prakash Kadayil
Michael Luxen
Sinan Cidem
Irem Isik Erol
Alexander Mödder



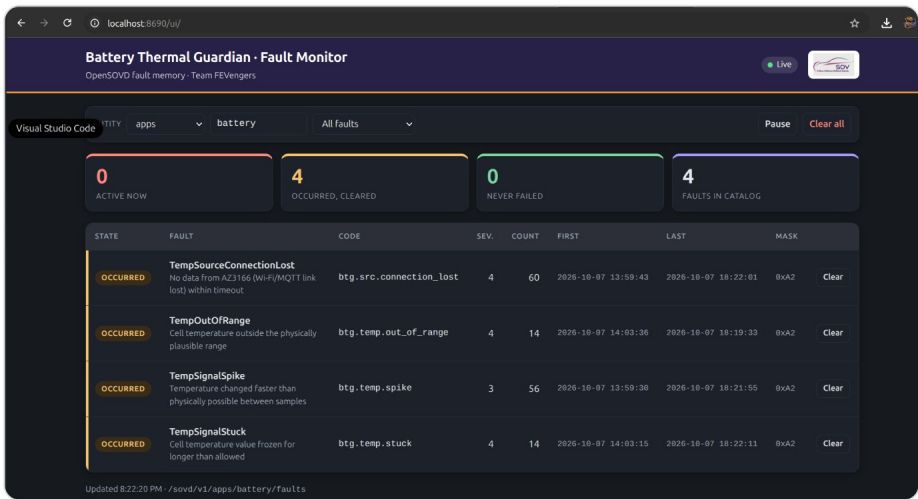
System architecture · from sensor to safety evidence



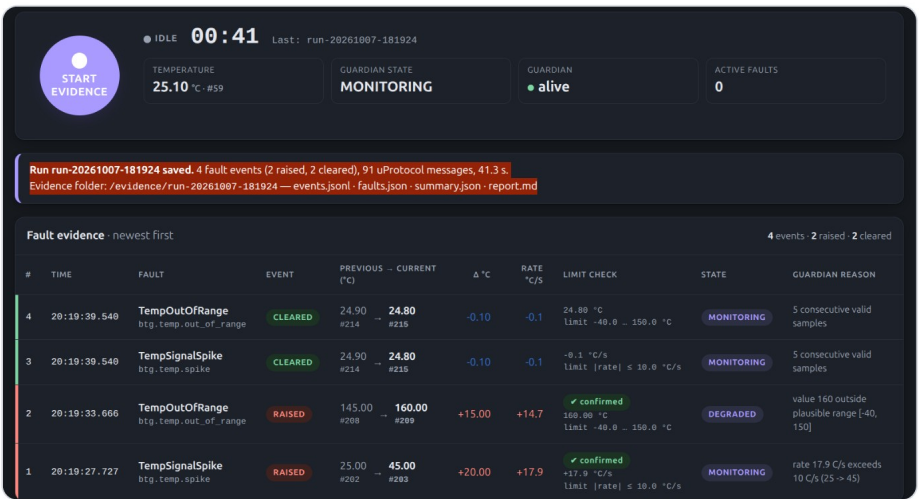
Eclipse technologies

- Eclipse ThreadX** iot.threadx
RTOS on the AZ3166 edge device · MQTT telemetry
- Eclipse Mosquitto** iot.mosquitto
MQTT broker for the board's Wi-Fi telemetry
- Eclipse uProtocol** automotive.uprotocol
Service contract: VSS in · Guardian events out
- Eclipse Zenoh** iot.zenoh
uProtocol transport between the containers
- Eclipse iceoryx2** technology.iceoryx
Zero-copy IPC: Guardian → DFM ↔ SOVD
- Eclipse OpenSOVD** automotive.opensovd
fault-lib, DFM and SOVD REST gateway
- Eclipse Ankaio's** automotive.ankaio's
Orchestration: desired state, restart, boot

Front end · two web UIs served from AutoSD, viewed on the host



Fault monitor · OpenSOVD gateway – DFM fault memory over SOVD REST /sovd/v1/apps/battery/faults, live, with clear



Evidence collector · independent judge – every fault and mitigation with previous → current value and its own limit check

AutoSD + Podman: everything is a container

- Built once on the laptop, shipped as OCI images, loaded on AutoSD unchanged
- Same image on Ubuntu and AutoSD – only run flags differ (SELinux, IPv4 Zenoh)
- Isolated by default; sharing is explicit: --ipc=host only for iceoryx2
- Swap or restart one service in seconds, the rest keeps running

How we worked: Scrum on GitHub

- Backlog and every task as GitHub Issues, worked in parallel by five people
- Branches USER/<name> → Integration → main
- Every merge through a reviewed pull request; conflicts solved on Integration
- Architecture first (Excalidraw), daily sync on Slack, **Release_1.0** tagged



Fault monitor · OpenSOVD gateway

Visual Studio Code

apps

battery

All faults

Pause

Clear all

0

ACTIVE NOW

4

OCCURRED, CLEARED

0

NEVER FAILED

4

FAULTS IN CATALOG

STATE	FAULT	CODE	SEV.	COUNT	FIRST	LAST	MASK	
OCCURRED	TempSourceConnectionLost No data from AZ3166 (Wi-Fi/MQTT link lost) within timeout	btg.src.connection_lost	4	60	2026-10-07 13:59:43	2026-10-07 18:22:01	0xA2	Clear
OCCURRED	TempOutOfRange Cell temperature outside the physically plausible range	btg.temp.out_of_range	4	14	2026-10-07 14:03:36	2026-10-07 18:19:33	0xA2	Clear
OCCURRED	TempSignalSpike Temperature changed faster than physically possible between samples	btg.temp.spike	3	56	2026-10-07 13:59:30	2026-10-07 18:21:55	0xA2	Clear
OCCURRED	TempSignalStuck Cell temperature value frozen for longer than allowed	btg.temp.stuck	4	14	2026-10-07 14:03:15	2026-10-07 18:22:11	0xA2	Clear

Updated 8:22:20 PM · /sovd/v1/apps/battery/faults

WHAT IT SHOWS

- The DFM fault memory, read every second over SOVD REST GET /sovd/v1/apps/battery/faults
- One state per fault: **ACTIVE**, **OCCURRED** or **OK**, from testFailed and testFailedSinceLastClear
- Occurrence counter, first and last occurrence, DTC status mask
- Clear one fault or all of them (SOVD DELETE) at the start of a test run

Served by **opensovd-gateway** on port **7690** at /ui/ · offline, no external resources

Evidence collector · independent judge

START EVIDENCE

● IDLE 00:41 Last: run-20261007-181924

TEMPERATURE 25.10 °C · #59

GUARDIAN STATE MONITORING

GUARDIAN ● alive

ACTIVE FAULTS 0

Run run-20261007-181924 saved. 4 fault events (2 raised, 2 cleared), 91 uProtocol messages, 41.3 s.
Evidence folder: /evidence/run-20261007-181924 — events.jsonl · faults.json · summary.json · report.md

Fault evidence · newest first 4 events · 2 raised · 2 cleared

#	TIME	FAULT	EVENT	PREVIOUS → CURRENT (°C)	Δ °C	RATE °C/s	LIMIT CHECK	STATE	GUARDIAN REASON
4	20:19:39.540	TempOutOfRange btg.temp.out_of_range	CLEARED	24.90 → 24.80 #214 → #215	-0.10	-0.1	24.80 °C limit -40.0 ... 150.0 °C	MONITORING	5 consecutive valid samples
3	20:19:39.540	TempSignalSpike btg.temp.spike	CLEARED	24.90 → 24.80 #214 → #215	-0.10	-0.1	-0.1 °C/s limit rate ≤ 10.0 °C/s	MONITORING	5 consecutive valid samples
2	20:19:33.666	TempOutOfRange btg.temp.out_of_range	RAISED	145.00 → 160.00 #208 → #209	+15.00	+14.7	160.00 °C limit -40.0 ... 150.0 °C	DEGRADED	value 160 outside plausible range [-40, 150]
1	20:19:27.727	TempSignalSpike btg.temp.spike	RAISED	25.00 → 45.00 #202 → #203	+20.00	+17.9	+17.9 °C/s limit rate ≤ 10.0 °C/s	MONITORING	rate 17.9 C/s exceeds 10 C/s (25 → 45)

WHAT IT SHOWS

- **Start / Stop evidence:** one recording run = one evidence folder
- Subscribes over uProtocol to Guardian events and the raw VSS samples – independent of the Guardian
- Every fault RAISED / CLEARED and mitigation REQUESTED / RELEASED with previous → current value, Δ and rate
- Its own limit check per fault: ✓ **confirmed** or ✗ **not seen**
- Writes events.jsonl · faults.json · summary.json · report.md, with SOVD before / after

Served by **evidence-collector** on port **7700** · runs as its own container



WHY CONTAINERS

One image.
Any host.
Same behaviour.

Six services, each its own OCI image. Built and tested on an Ubuntu laptop, run unchanged on AutoSD in a QEMU VM.

Build once, run anywhere

Rust compiled inside the build container – no toolchain on AutoSD, no "works on my machine"

Isolated by default, shared by choice

Only the iceoryx2 users share IPC and /dev/shm; everything else stays apart

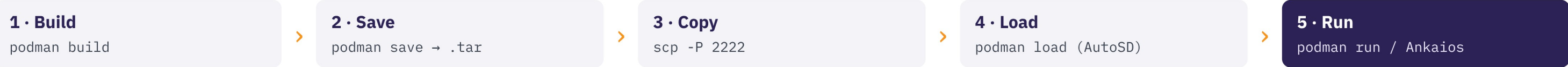
Replace one service in seconds

Rebuild, load, recreate one container – the rest of the chain keeps running

Orchestrated by Ankaaios

Desired state in one manifest: start at boot, restart on failure, start order by dependency

FROM LAPTOP TO AUTOSD



THE SIX CONTAINERS ON AUTOSD

CONTAINER	ROLE	PORT	ICEORYX2	ZENOH
mqtt-broker	Eclipse Mosquitto, receives the board	1883	–	–
vss-publisher	MQTT → VSS → uProtocol	–	–	✓
guardian	Battery Thermal Guardian	–	✓	✓
dfm	Diagnostic Fault Manager	–	✓	–
opensovd-gateway	SOVD REST + fault monitor	7690	✓	–
evidence-collector	Independent judge + web UI	7700	–	✓

WHAT AUTOSD ASKED OF US

SELinux enforcing
shared mounts need --security-opt label=disable

IPv6 disabled
Zenoh gets an IPv4-only listener tcp/0.0.0.0:0

/tmp is a tmpfs
/tmp/iceoryx2 is recreated after every reboot

Shared memory across containers
--ipc=host --pid=host + /dev/shm for every iceoryx2 user



THE TEAM · FEV.io × FEV TÜRKIYE

NAME	GITHUB	EXPERIENCE	ROLE
Ashwin Prakash Kadayil	Kadayil	Embedded systems, HPC, software architecture	Hacking
Michael Luxen	FEVLuM	In-vehicle networks, diagnostics, embedded SW	Hacking
Sinan Cidem	CidemSinan	Embedded engineering, Ankaïos, AUTOSAR	Hacking
Irem Isik Erol	isik-i	VCU integration, AUTOSAR, DevOps	Hacking
Alexander Mödder	MoedderAlex	Simulation, network communication, DevOps	Project management

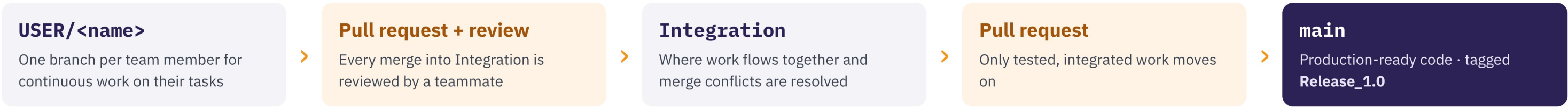
DECISION MAKING

Popular vote.
Five people cannot tie.

TOOLS

- GitHub · code, issues, PRs
- Excalidraw · architecture
- Slack · daily work
- WhatsApp · off-site

BRANCHING STRATEGY · EVERY PUSH TO A SHARED BRANCH GOES THROUGH A PULL REQUEST



BACKLOG

GitHub Issues

Every task and work item is an issue – the shared backlog the five of us pulled from in parallel.

QUALITY

Reviewed pull requests

Code review on every merge into Integration; unit tests for as many components as possible.

PLANNING

Architecture first

A joint brainstorm on Excalidraw exposed missing APIs and dependencies early, so exploration could run in parallel.

DOCUMENTATION

Shared and up to date

Each member documents their own work with every feature or change, so the docs stay current.